

Safety Code Quick Reference - ASME A17.1 / CAN/CSA B44

Rocket Elevators - Ontario Fleet Technical Reference

Overview

This document summarises key ASME A17.1-2019 / CAN/CSA B44-19 requirements relevant to Ontario elevator compliance inspections. Always consult the full standard for enforcement decisions. References are to section numbers.

Load and Speed Requirements

2.16.1 - Rated load: Passenger elevators must be designed for 272 kg/m² of net platform area, not less than 454 kg total.

2.16.3 - Contract speed must not exceed 10% above rated speed under no-load.

3.19.1 - Relief valve setting: not more than 125% of working pressure.

Safety Device Tests (Periodic)

8.6.4.1 - Safety and governor test: required every five years for traction elevators; every hydraulic test cycle for hydraulic elevators.

8.6.5 - Buffer test: required at full contract speed with 100% load (or counterweight buffer test with empty car at contract speed).

8.11.1 - Oil buffer recharge: within four hours of test completion.

Door Locking and Closing Force

2.1.3.3 - Kinetic energy of closing door must not exceed 3.5 J.

2.1.3.4 - Maximum closing force at 25 mm from fully closed: 133 N.

2.12.5 - Hoistway access doors must be self-closing and self-locking; can only be opened from the hoistway side by an authorized person.

Fire Service Operation

2.27.2 - Phase I recall: on smoke detector activation, elevator must return to primary landing (or alternate) within 60 seconds. Doors must open and remain open. Car must not respond to car calls.

2.27.3 - Phase II operation: firefighter toggle enables manual operation. Door open button must keep doors open. Test annually.

Clearances - Pit and Overhead

2.2.1 - Pit depth: minimum 1050 mm below lowest landing sill.

2.2.2 - Overhead: minimum clearance from top of car to overhead structure when car is at top landing with full-speed safety test in effect.

2.2.5 - No projections in pit other than buffers, guide rails, hydraulic cylinders, and pit equipment.

Maintenance Requirement Summary

8.6.1 - Periodic inspections: every 12 months for all elevating devices.

8.6.2 - Follow-up inspections: required within 60 days of a deficiency order.

8.11 - Maintenance records must be retained at the premises for 3 years and made available to TSSA inspectors on request.